

Programming Scalable Systems

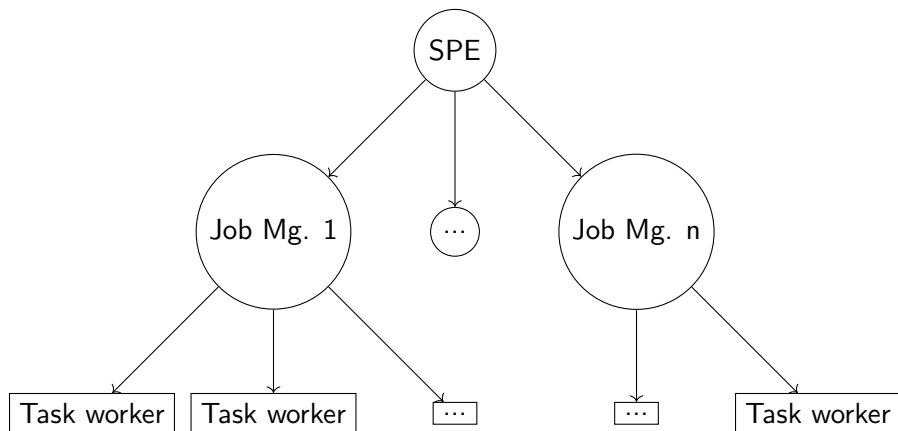
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Design



Messages SPE

From	To	Message
Client	SPE	start
Client	SPE	submit_job
Client	SPE	start_job
SPE	Job manager	start
SPE	Job manager	start_one_task

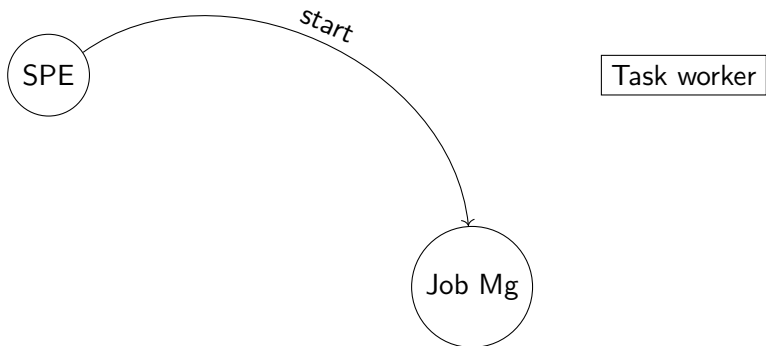
Messages Job Manager

From	To	Message
Job Manager	SPE	existing_worker
Job Manager	SPE	one_task_finished
Job Manager	SPE	job_finished
Job Manager	Task worker	start

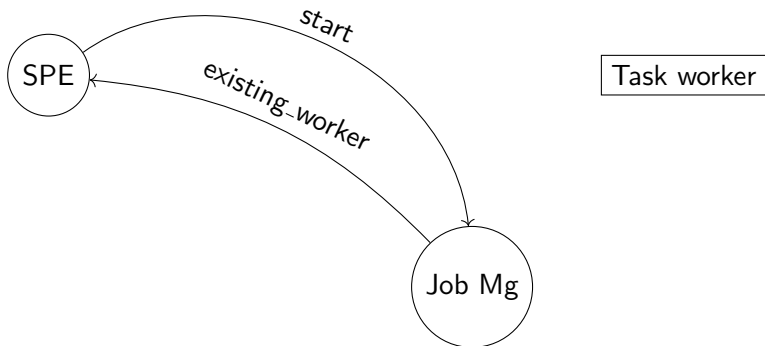
Task worker

From	To	Message
Task worker	Job Manager	task_finished

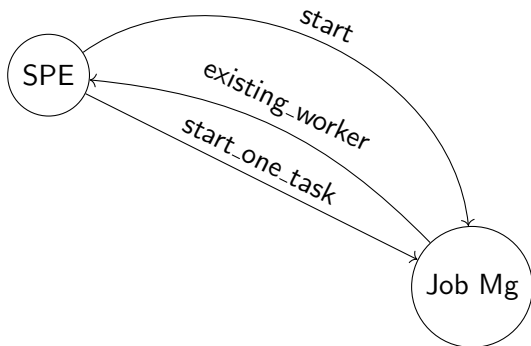
Summary of messages



Summary of messages

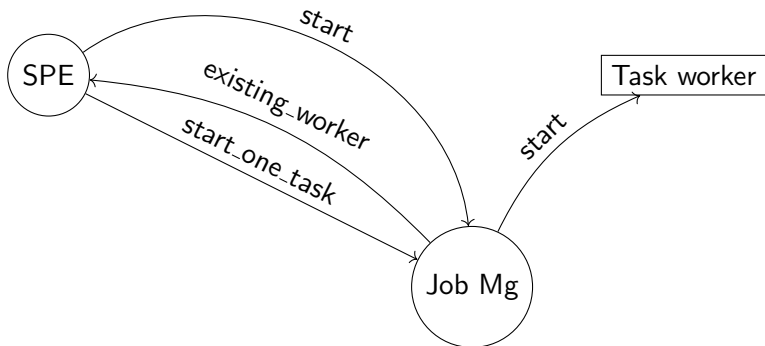


Summary of messages

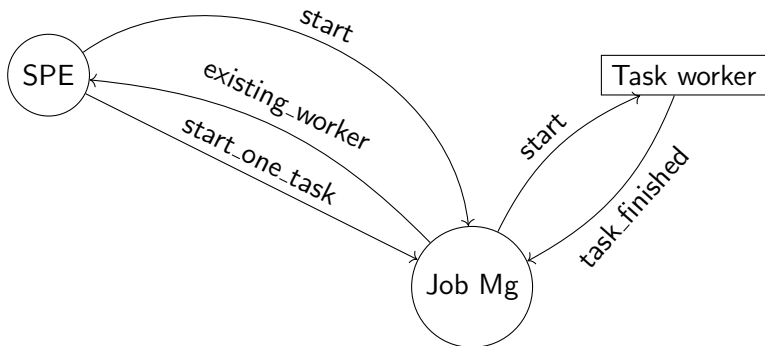


Task worker

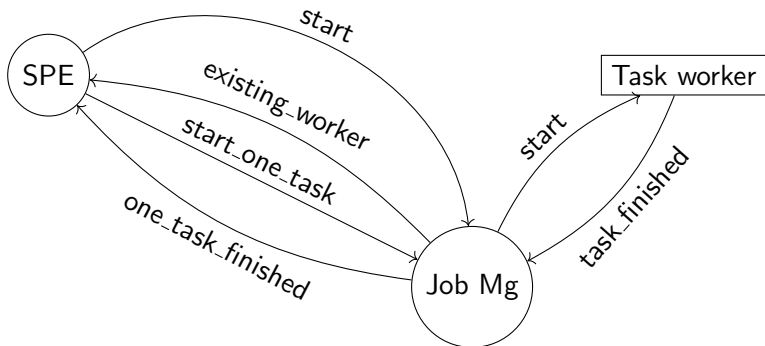
Summary of messages



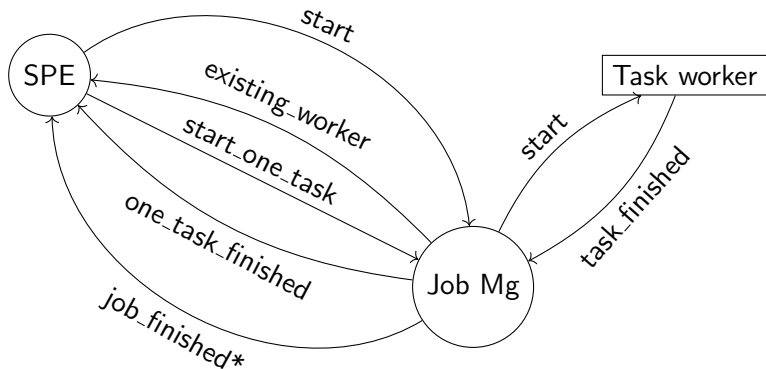
Summary of messages



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Implementation details

Element	Implementation
SPE	GenServer
Job manager	GenServer
Task worker	Task

State SPE

- num_workers (int)
- current_workers (int)
- pid_queued_jobs (list of PIDs)
- jobs (map)
- next_unique_id (int)

State Job Manager

- job_id (int)
- name (string)
- server_pid (PID)
- queued (list of tasks)
- blocked (list of tasks)
- running (map of tasks)
- status (atom)
- results (map of task-results)

Task implementation

```
Task.start(  
    fn ->  
        run(name, fun, input, timeout, caller)  
    end  
)
```

```
defp execute_with_timeout(function, input, ms)  
defp execute_without_timeout(function, input)  
defp safe_execution(function, input)
```